

Recap HW2 posted
Midterm Nov 18

- Existence & uniqueness proof
uses contraction mapping principle.
- Solutions & stability of linear equations

Contraction maps

M : Complete metric space

A map $F: M \rightarrow M$ is called a contraction if

$$\rightarrow d(F(x), F(y)) \leq \lambda d(x, y)$$

for all $x, y \in M$ and some $\lambda \in (0, 1)$.

Let $x \neq y$

n :

$$d(F(x), F(y)) \leq \lambda d(x, y)$$

$$\rightarrow d(F^t(x), F^t(y)) \leq \lambda^t d(x, y) \quad (*)$$

$t \geq 1$.

$$\rightarrow t \in \mathbb{Z}^+$$

$$\rightarrow F^0 = Id$$

$$d(F^2(x), F^2(y)) \leq \lambda d(F(x), F(y)) \leq \lambda \cdot \lambda d(x, y)$$

as $t \rightarrow \infty$, orbits converge.

$\downarrow$
 $\{F^t(x)\}_{t \in \mathbb{Z}^+}$ is called an orbit of x .

Appear in PDEs, inverse function theorem, existence & uniqueness proofs
Optimization, Newton's method...

A contraction map asymptotically goes to a fixed point.

From $(*)$, WKT if an orbit converges to a point, say x^* , then any orbit converges to x^* .

(Fixed point x^* is unique.

$$F(x^*) = x^*)$$

Want to show that a fixed point x^* exists.

Orbit of an $x \in M$ is a Cauchy sequence.

$\{F^t(x)\}_{t \in \mathbb{Z}^+}$ is Cauchy.

$m \geq t$

$$d(F^m(x), F^t(x)) \leq \sum_{\tau=t}^{m-1} d(F^{\tau+1}(x), F^{\tau}(x)) \quad (A)$$

(triangle inequality)

$$d(x, z) \leq d(x, y) + d(y, z)$$

But, WKT,

$$d(F^{\tau+1}(x), F^{\tau}(x)) \leq \lambda^{\tau} d(F(x), x)$$

(by applying $(*)$ to $x = x, y = F(x)$)

Putting this back in (A) , $m-1$

$$d(F^m(x), F^t(x)) \leq \sum_{\tau=t}^{m-1} \lambda^{\tau} d(F(x), x)$$

$$\leq \frac{\lambda^t d(F(x), x)}{(1-\lambda)} \quad (B)$$

$< \epsilon$

Hence $\{F^t(x)\}$ is a Cauchy sequence.

$\Rightarrow x^*$ exists & is unique.

In (B) , take $m \rightarrow \infty$,

$$d(x^*, F^t(x)) \leq \lambda^t C$$

The orbit of any x approaches x^* exponentially fast.

For any $x \in M$, if $\|\nabla F(x)\| < 1$,
then F is a contraction $\sup_{x \in M} \|\nabla F(x)\| < 1$

$$\|F(x) - F(y)\| \leq \|\nabla F(\xi)\| \|x - y\|$$

(MVT).

F is often hidden!

$$x_{t+1} = A x_t$$

$$x_t \in \mathbb{R}^n$$

$$A \in \mathbb{R}^{n \times n}$$

$F(x) = Ax$ is a contraction
when maximal absolute eigenvalue of

$$A = \lambda < 1$$

$$\|A^t x\| \leq \lambda^t \|x\| \rightarrow 0$$

as $t \rightarrow \infty$.

$$\frac{dx}{dt} = Ax$$

in Lec 2, wrote down solutions using
eigendecomposition of A

Jordan form of A .

$$\begin{aligned} x(t) &= e^{tA} x(0) \\ &= P^{-1} e^{t\Lambda} P x(0). \end{aligned}$$

Is a contraction when $\operatorname{Re}(\lambda) < 0$
for all eigenvalues λ of A .

Linear stability analysis

What happens to infinitesimal perturbations of orbits?

$x \in M$ (phase space)
compact set in $\mathbb{R}^n$

$$\frac{dx}{dt} = V(x)$$

using "flow" notation

$$\frac{d\varphi^t(x)}{dt} = \frac{V(\varphi^t(x))}{t \in (\text{interval where}) \text{ solutions exist on } M.}$$

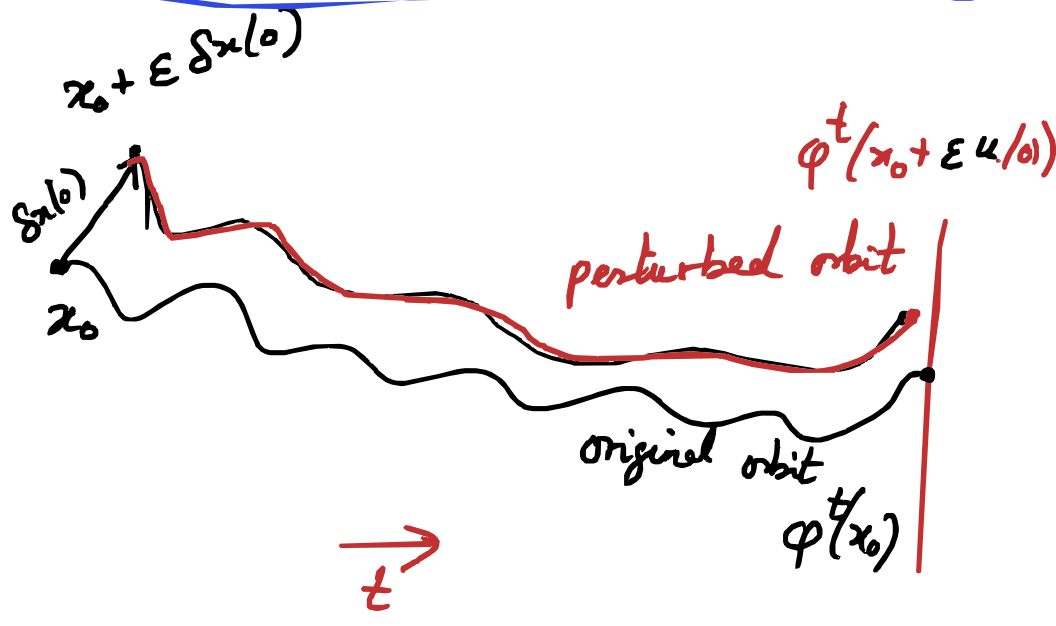
x : initial condition

$\{\varphi^t(x)\}_{t \in [0, \epsilon]}$ orbit of x

$u(0)$: infinitesimal perturbation to $x_0 \in M$.

$u(t)$?

$$u(t) = \lim_{\epsilon \rightarrow 0} \frac{\varphi^t(x_0 + \epsilon u(0)) - \varphi^t(x_0)}{\epsilon} \quad \text{(A)}$$



$$\frac{d u(t)}{dt} = \lim_{\epsilon \rightarrow 0} \frac{1}{\epsilon} \left(\frac{d\varphi^t(x_0 + \epsilon u(0))}{dt} - \frac{d\varphi^t(x_0)}{dt} \right)$$

$$= \lim_{\epsilon \rightarrow 0} \frac{1}{\epsilon} \left(V(\varphi^t(x_0 + \epsilon u(0))) - V(\varphi^t(x_0)) \right)$$

(from ODE)

$$= \lim_{\epsilon \rightarrow 0} \frac{1}{\epsilon} \left(V(\varphi^t(x_0)) + \epsilon \nabla V(\varphi^t(x_0)) u(t) - V(\varphi^t(x_0)) \right)$$

[$V \in C^1(M)$ and definition of $\delta x(t)$]

$$\varphi^t(x_0 + \epsilon u(0)) = \varphi^t(x_0) + \epsilon u(t) + O(\epsilon^2)$$

from (A)

$$\frac{d u(t)}{dt} = \nabla V(\varphi^t(x_0)) u(t)$$

(C)

① The orbit $\varphi^t(x_0)$ is fixed.

② The ODE that describes $\delta x(t)$: infinitesimal perturbation at time t to $\varphi^t(x_0)$ is LINEAR

$$\frac{dy}{dt} = Y(t, y)$$

③ The ODE that describes $\delta x(t)$ is non-autonomous since the matrix $\nabla V(\varphi^t(x_0)) \in \mathbb{R}^{n \times n}$ is changing with time

Linear stability analysis around a fixed point

$u(t)$: inf. perturb. to $\varphi^t(x_0)$

$x_0 \equiv x^*$ a fixed point for φ^t

$$\varphi^t(x^*) = x^* \text{ for all } t$$

(defn of fixed point).

$$\frac{du(t)}{dt} = \nabla V(x^*) u(t)$$

(evolution of inf. perturbation to
fixed pt x^*)

Ex: put $\varphi^t(x_0) = x^*$ in the
derivation of $\frac{du(t)}{dt} = \nabla V(\varphi^t(x_0)) u(t)$

If
 $\text{Re}(\lambda) < 0$ for all eigenvalues
 λ of $\nabla V(x^*)$, then,
 $u(t) \rightarrow 0 \in \mathbb{R}^n$ as $t \rightarrow \infty$.

$$u(t) = e^{tA} u(0)$$

$$\text{where } A = \nabla V(x^*)$$

Holds "locally" around fixed pts of
any nonlinear flow $\varphi^t(x)$.

Next time:

$$y(t) = \varphi^t(x) - x_*$$

when x is near x_* .